

AMENDMENT NO. 2 OCTOBER 1978

TO

IS : 2825 - 1969 CODE FOR UNFIRED PRESSURE VESSELS

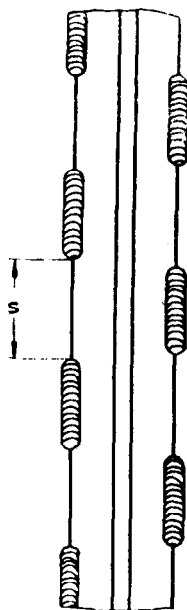
Alterations

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| <p><i>Sl No.</i></p> <ol style="list-style-type: none"> 1. (Page 6, clause 1.3.1.1, para 2) — Substitute the following for the existing paragraph:
 ‘All welded joints of Class 1 vessels shall meet the requirements stipulated in col 3 of Table 1.1. All butt joints shall be fully radiographed. Circumferential butt joints in nozzles, branches and sumps not exceeding 170 mm outside diameter or 19 mm wall thickness need not be radiographed (see 8.7.1.1) except for vessels that are to contain lethal* or toxic substances.’ 2. (Page 6, clause 1.3.1.1, paras 4 and 5) — Substitute the following for the existing paragraphs:
 ‘a) <i>Category A</i> — Longitudinal welded joints within the main shell, communicating chambers†, transitions in diameter, or nozzles; any welded joint within a sphere, within a formed or flat head, or within the side plates of a flat sided vessel; circumferential welded joints connecting hemispherical heads to main shells, to transitions in diameters, to nozzles, or to communicating chambers†.
 b) <i>Category B</i> — Circumferential welded joints within the main shell, communicating chambers†, nozzles or transitions in diameter including joints between the transition and a cylinder at either the large or small end; circumferential welded joints connecting formed heads other than hemispherical to main shells, to transitions in diameter, to nozzles, or to communicating chambers†.’ 3. [Page 8, Table 1.1, against <i>Sl No.</i> 3(b), col (5), (6) and (7)] — Substitute the following for the existing matter in each of these columns:
 <div style="margin-left: 40px;"> ‘Maximum thickness 16 mm before corrosion allowance is added, and 18 mm after adding corrosion allowance’ </div> 4. (Page 11, clause 2.2.4) — Substitute the following for the existing clause:
 ‘2.2.4 <i>Bearing Stress</i> — The maximum permissible bearing stress shall not exceed the allowable stress value.’ | <p><i>Sl No.</i></p> <ol style="list-style-type: none"> 5. [Page 11, clause 3.1.3.1(c)] — Substitute the following for the existing text:
 ‘c) Wind loading in combination with other loadings (see C-4.1.3),
 d) Seismic loads (see IS : 1893-1975*)’. 6. (Page 11, foot-note) — Add the following new foot-note:
 ‘*Criteria for earthquake resistant design of structures (second revision).’ 7. (Page 12, clause 3.2.2) — Add the following note after the clause:
 ‘NOTE — All calculations in this Code are in corroded condition unless otherwise specified.’ 8. (Page 12, clause 3.2.3, para 2) — Add the following at the end of the paragraph:
 ‘In such cases tell-tale holes, suitably disposed, shall be provided on the vessel wall so that easy detection of leakage in welds in lining is made possible. They shall have a depth not less than 80 percent of the thickness required for a seamless shell of like dimensions or they may extend to the lining.’ 9. (Page 13, clause 3.3.2.1, definition of symbol <i>E</i>) — Delete the words ‘see Tables 3.1, 3.2, 3.3 and 3.4’. 10. (Page 13, clause 3.3.2.4, definition of symbols <i>E</i>, <i>E_a</i>) — Delete the words ‘(see Tables 3.1, 3.2, 3.3 and 3.4)’. 11. (Page 14, Tables 3.1, 3.2, 3.3 and 3.4) — Delete. 12. (Page 15, clauses 3.3.3 and 3.3.3.1) — Substitute the following for the existing clauses:
 ‘3.3.3 <i>Shells Subjected to External Pressure</i>
 3.3.3.1 The thickness of shells subjected to external pressure shall be calculated by the method given in Appendix F.’ 13. [Page 15, clauses 3.3.3.2, 3.3.3.3, 3.3.3.4(a) and 3.3.3.4(b)] — Delete these clauses. 14. (Page 16, new clause 3.3.3.2) — Add the following new clause 3.3.3.2 at the top of the page:
 ‘3.3.3.2 <i>Stiffening rings</i> — Stiffening rings are generally used with cylindrical shells subjected to external pressure. They extend around the circumference of the shell and may be located on the inside or the outside of the shell.’ |
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No.

15. [Page 16, clause 3.3.3.4(c)] — Re-number the clause as 3.3.3.2(a).
16. [Page 17, clause 3.3.3.4(d)] — Re-number the clause as 3.3.3.2(b).
17. [Page 17, clause 3.3.3.4(d) (1), renumbered as clause 3.3.3.2(b) (1)] — Substitute the following for the existing text:

'1) the total length of intermittent welding on each side of the stiffening ring shall be not less than one-half the outside circumference of the vessel, for stiffening rings situated on the outside. However, at any location of the stiffening ring there shall be welding between the shell and at least one side of the stiffening ring. The maximum unwelded length s on any side shall not exceed 8 times the vessel thickness for external rings and 12 times the vessel thickness for internal rings as shown below:



18. [Page 17, clause 3.3.3.4(e)] — Renumber the clause as 3.3.3.3.
19. (Page 17, clause 3.3.3.5) — Renumber the clause as 3.3.3.4.
20. (Page 30, clause 3.8.5.1) — Substitute the following for the existing clause:

'3.8.5.1 General — At all planes through the axis of the opening normal to the vessel surface the cross-sectional area requirements for compensation as calculated below shall be satisfied:

Material in added compensation, or in a branch, should have similar mechanical and physical properties to that in the

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No.

vessel shell or end. It is recommended that compensation or fittings be made normally from material having an allowable stress not less than 75 percent of the allowable stress for the material in the shell or end. Where material having a lower allowable stress than that of the vessel shell or end is taken as compensation its effective area shall be assumed to be reduced in the ratio of the allowable stresses at the design temperature. No credit shall be taken for the additional strength of material having a higher stress value than that in the shell or end of the vessel.

Material added for compensation shall have approximately same coefficient of thermal expansion (say 85 percent). The thickness of compensating pad shall be limited to the nominal thickness of shell or head as relevant. Any area contributing to compensation shall be attached to shell or head by full penetration welds.'

21. (Page 35, informal table under clause 3.8.9.1) — Substitute the following for the existing table:

Branch Nominal Size	Minimum Thickness
mm	mm
15	2.4
20	2.4
25	2.7
32	3.1
40	3.1
50	3.6
65	3.9
80	4.7
100	5.4
125	5.4
150	6.2
200	6.9
250	8.0
300	8.0
350	8.8
400	8.8
450	8.8
500	10.0
600	10.0

NOTE — The value given in the table is to be increased by the amount of any required corrosion allowance.'

22. (Page 43, clause 4.1.3) — Substitute the following for the existing clause:

'4.1.3 Hub flanges shall not be made by machining the hub directly from plate materials.'

23. (Pages 56, 61, 62, various sub-clauses of clause 5) — Substitute 'device' for 'valve' and 'devices' for 'valves' wherever it appears.

24. (Page 61, clause 5.4.1) — Substitute the following for the existing clause:

Sl
No.

'5.4.1 The total capacity of the relief device or devices fitted to any vessel or system of vessels shall be sufficient to discharge the maximum quantity of fluid, liquid or gas, that can be generated or supplied without permitting a rise in vessel pressure of more than 10 percent above the maximum working pressure when the relief devices are discharging.'

25. (Page 65, clause 6.2.1.3) — Substitute the following for the existing clause:

'6.2.1.3 The weld procedure and welders shall be qualified for the type of welding concerned in conformity with the weld procedure and welders' performance qualifications (see 7.1 and 7.2).'

26. (Pages 65 and 66, clause 6.3.2) — Substitute the following for the existing clause:

'6.3.2 In the design of all details the aim shall be to avoid disturbances in the flow of the lines of force, in particular in constructions subjected to fatigue stresses. Holes and openings shall not be positioned on or in the heat affected zones of welded joints. Where such openings are unavoidable, such holes can be located on circumferential joints provided the main seams are ground flush 75 mm on either side and radiography requirements given in 6.3.2(a) are met. However openings with added reinforcements should not have the hole cut on the weld seam.

- a) The circumferential seam shall be radiographed for a total length of three times the diameter of opening with the centre of whole at mid length. Defects that are completely removed in cutting the hole shall not be considered in judging the acceptability of the weld.'

27. (Page 66, clause 6.3.2.1, second sentence) — Substitute the following for the existing sentence:

'Joints between cylindrical shells and domed end plates shall not be located in the curved part of the domed end except for hemispherical ends (see Table 6.2).'

28. (Page 67, clause 6.4.2.5) — Substitute the following for the existing clause:

'6.4.2.5 Edges which have been flame-cut by hand shall be cut by machining or chipping for a distance not less than 1.5 mm.'

29. [Page 71, clause 6.6.2(b)(2), line 3] — Substitute '5 mm' for '4 mm'.

30. (Page 74, clause 6.7.16, last sentence) — Substitute the following for the existing sentence:

Sl
No.

'In the case of light duty vessels the reinforcement shall be not more than 3 mm for thickness less than 10 mm, and 5 mm for thickness from 10 to 16 mm.'

31. [Page 76, Table 6.3, col (3), para 2, line 4] — Insert the following between the words 'preheating' and 'for':

'up to a minimum temperature of 150°C'

32. (Page 76, Table 6.3, Note 2, line 1) — Substitute '16 mm' for '15 mm'.

33. (Page 76, Table 6.3, Note 3, line 1) — Substitute '0°C' for '-20°C'.

34. [Page 81, clause 7.1.3.4(e)] — Substitute the following for the existing clause:

'e) if in gas shielded metal arc (MIG) or tungsten arc (TIG) welding or submerged arc welding a change is made from multiple pass welding per side to single pass welding per side when notch toughness is a requirement.'

35. [Page 81, clause 7.1.3.4(h)] — Substitute the following for the existing clause:

'h) if in gas shielded metal arc welding (MIG) or gas shielded tungsten arc welding (TIG) a change is made in the composition of the gas and a change in the electrode from one type to another or from non-consumable electrode to consumable electrode and *vice versa*, or an increase of 25 percent or more or a decrease of 10 percent or more in the rate of flow of shielding gas.'

36. (Page 81, clause 7.1.3.5) — Add the following new paragraph after the clause:

'A change from downward to upward or *vice versa* in the progression for any pass of a vertical weld excepting the root pass which will be removed completely while preparing the second side and the cover pass, if notch toughness is a requirement, calls for requalification.'

37. (Page 81, clause 7.1.4, first sentence) — Add the following new matter after the first sentence:

'However, for temperatures less than 0°C the qualification shall be valid for thickness from 1 to 1.1 times the thickness of test piece.'

38. [Page 82, clause 7.1.5(b), lines 3 and 4] — Delete the words 'one above the other,'

39. [Page 82, clause 7.1.5(c), lines 3, 4 and 5] — Delete the words 'and where the

Sl
No.

thickness of the plate permits, one shall be above the other'.

40. [Page 82, clause **7.1.5(c)(1)**, line 2] — Substitute '20 mm' for '30 mm'.
41. (Page 87, Table 7.4, column of maximum range of thickness qualified by test plate corresponding to thickness of test plate 't 20 and over') — Substitute '2 t' for '1.5 t'.
42. (Page 88, clause **7.2.6.4**) — Add the following new clause after this clause:
'**7.2.6.5** Radiography is acceptable in place of mechanical testing for manual metal arc or gas shielded tungsten arc or combination processes.'
43. (Page 95, clause **8.4.2**, line 1) — Delete the words ' in mild or low-alloy steels ' .
44. [Page 97, clause **8.5.1.3(c)**, para 1, lines 3, 4 and 5] — Delete the words ' and where the thickness of plate permits, one shall be above the other '.
45. [Page 97, clause **8.5.1.3(c)**, para 2, line 2] — Substitute '20 mm' for '30 mm'.
46. [Page 97, clause **8.5.2.2(b)**, first and second sentence] — Substitute the following for the existing sentences:
'Two bend test specimens, one for direct and one for reverse bending to be taken transversely to the weld. Where the plate thickness exceeds 20 mm, face bend and root bend tests may be substituted by side bend tests.'
47. (Page 100, clause **8.5.9.2**, line 2) — Substitute '20 mm' for '30 mm'.
48. (Page 100, clause **8.5.9.3**, lines 2, 3 and 7) — In each of the lines, substitute '20 mm' for '30 mm'.
49. (Page 101, clause **8.5.9.6**) — Add the following new matter in the beginning:
'When specifically asked for by the customer'.
50. [Page 103, clause **8.6.8.1(a)**] — Add the following new matter at the end:
'For the transverse tensile test if the

Sl
No.

result exceeds 95 percent of the specified minimum value with fracture occurring in base metal, the test is acceptable.'

51. (Page 103, clause **8.7.1.2**) — Delete.
52. (Page 104, clause **8.7.4**, last sentence) — Substitute the following for the existing sentence:
'When image quality indicator of the wire-type is used, the smallest diameter wire which can be seen in the radiograph shall have a diameter not greater than 1.5 percent of the weld metal thickness if the weld metal thickness is between 10 and 50 mm inclusive, and not greater than 1.25 percent of the weld metal thickness if the thickness is between 50 mm and 200 mm. In the case of plate type penetrameters, the radiographic examination shall be capable of revealing a difference in metal thickness equal to not more than 2 percent of the thickness of weld under examination.'
53. (Page 109, clause **8.7.10.2**, para 1) — Delete and renumber the second paragraph as '**8.7.10.2**'.
54. (Page 110, clause **8.7.11**, first sentence) — Substitute the following for the existing sentence:
'When special conditions make it expedient, radiography as specified in **8.7.1** and **8.7.2** may be replaced by ultrasonic testing supplemented by dye-penetrant or magnetic particles inspection subject to the previous consent of the inspecting authority and on the condition that such testing methods may be considered to render an equally safe evaluation of the quality of welding work.'
55. (Page 178, clause **D-6.5**) — Add the following new clause after this clause:
'**D-6.6** All nozzles to shell attachments and nozzles to flange attachments shall be full penetration welds.'